

Bio-Inspired Design and Finite Element Simulation of a Mechanical Tremor-Stabilising Spoon

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Abstract

A lot of people have hand tremors - mostly those with essential tremor and Parkinson's like essential tremor and Parkinson's diseases. Hand tremors have made simple tasks like eating extremely difficult. There are already available electronic devices such as stabilizing spoons on the market, but they are very expensive and require constant recharging. This research paper introduces a novel concept of tremor stabilising spoon with a purely mechanical design, which is based on natural phenomena. In order to minimize the impact of tremor, the bowl of the spoon is designed with a protective layer like silkworm cocoon, while the handle is based on bamboo-like segmented joints to provide greater durability. Moreover, the neck area includes a mechanism with an orthogonal spring structure like a shock-absorbing hyoid bone in woodpeckers' skull. This innovative design was modelled in SolidWorks and subjected to static structural, modal frequency, and thermal analysis. Simulation results show that the first natural frequency of 58.52 Hz is significantly higher than typical frequency ranges of hand tremor oscillations (4-12 Hz), thus eliminating any possibility of resonant amplification. Maximum stress of 13.8 MPa caused by a 1 N load is significantly lower than the yield point of AISI 304 stainless steel (207 MPa). Furthermore, there is effective insulation of heat transfer between bowl and handle areas. Finally, the designed spoon weighs 87.89 grams, thus reaching a targeted weight limit for comfortable operation.

Keywords

Engineering, Biomedical Engineering, biomimicry, tremor stabilisation, assistive device

Introduction

Essential tremor is known to be the most common pathological type of tremor seen in adults, with prevalence of around 0.9% in the general population.¹ Hand tremors usually occur at

frequencies in the range of 4–12 Hz with amplitudes from a few millimetres to several centimeters.² Eating for people suffering from these problems becomes quite challenging due to the uncontrolled shaking which prevents them from picking up food from a plate and putting it in their mouth without spilling. This situation leads to embarrassment and social alienation.

Weighted utensils, swivel spoons, and electronic stabilizing gadgets, such as Liftware Steady, are currently used to help tremors when eating. They consist of electronic sensors and motors which can sense hand tremors and move the tool oppositely to it, thus achieving 70% tremor dampening. Nevertheless, the problem with electronic instruments is their high cost and the need to recharge them. Moreover, some people might be unable to use electronic utensils because of their price in their countries of origin.

Research has proven that simply weighting utensils does not help in decreasing tremors and even may cause muscle exhaustion.³ Thus, it requires a more advanced approach. In this research project, we try to implement the idea of designing a gadget using mechanical principles inspired by nature and specifically the principle of orthogonal spring vibration isolation to achieve stabilizing effect without using electronics.

Nature has created many effective methods for absorbing shocks and vibration damping. Woodpeckers beat tree surfaces with decelerations higher than 1200g, and yet they do not get any brain damages hitting it with their beaks up to 18 to 22 times a second.⁴

Their hyoid bone, which wraps around the skull in a spring-like configuration, plays a crucial role in this shock absorption. Similarly, bamboo's nodal structure provides exceptional strength-to-weight ratio through its hierarchical fibre reinforcement schemes.⁵ The silkworm cocoon has developed a layered structure that gives good protection from environmental hazards despite its light weight.⁶

This research focuses on developing a purely mechanical tremor stabilization spoon based on the biological inspiration of nature's designs. The research intends to achieve four major objectives, which include: (1) design a spoon based on bio-mimicry principles, (2) generate an advanced computer-aided design (CAD) in SolidWorks software, (3) analyze the design using finite elements simulations in terms of static, frequency, and thermal aspects, and (4) verify the design for practicality and safety.

Hand Tremors and Existing Solutions

Tremors are described by the International Parkinson and Movement Disorder Society as involuntary rhythmic oscillations of a body segment. The primary causes of tremors in the hands are essential tremor and Parkinson's disease. Studies have demonstrated that the frequencies of essential tremors fall within the range of 4 to 12 Hz, whereas voluntary actions of the hand take place at less than 3 Hz.⁷

A number of gadgets have been designed for feeding problems. Weighted utensils attempt to use inertia to resist tremor-induced movements. However, Meshack and Norman found no significant improvement with static weight addition; in fact, tremor was often worse with weight due to increased muscle fatigue.⁸ Rocon et al. developed a wearable orthosis applying viscosity and inertia to the arm, achieving 40% power reduction through mechanical loading.⁹ This study validated that changing the impedance characteristics of the limb directly affects tremor.

Electronic solutions like the Liftware Steady use Active Cancellation of Tremor technology, detecting tremor direction and moving the utensil attachment in the opposite direction. Clinical trials showed 71% to 76% tremor reduction across eating tasks.² However, the cost and maintenance requirements limit accessibility. This gap motivated the exploration of purely mechanical alternatives.

Bio-Inspired Design Principles

Biomimicry offers valuable insights for engineering design. The woodpecker's ability to withstand repeated high-impact forces has inspired numerous shock-absorbing systems. Yoon and Park analysed the head structures of the golden-fronted woodpecker using X-ray computed tomography and developed a bio-inspired shock-absorbing system achieving a failure rate of only 0.7% at 60,000 g compared to 26.4% for conventional methods.⁴ The hyoid bone, which wraps around the skull, distributes mechanical excitations and provides spring-like energy absorption.

Bamboo's mechanical properties have attracted significant research interest. Chen et al. identified three hierarchical fibre reinforcement schemes in bamboo nodes: spatially tightened interlocking, triaxial interconnected scaffolding, and isotropic intertwining.⁵ These schemes provide exceptional structural stability. The nodal regions act as stress dissipation points, improving lateral strength and energy absorption. The tensile strength-to-specific weight ratio of bamboo is approximately six times greater than that of steel.

Silk worm cocoons also constitute another great instance of biological armor. According to Zhang et al., wild silk cocoon displays an exceptionally good capacity for toughness, heat buffering, moisture transport in one direction, and UV stability.⁶ The multi-layered nonwoven armor has evolved to offer effective protection without any significant weight. This formed the basis of development of the spoon bowl in the current study.

Tuned Mass Damper Principles

The tuned mass damper is an apparatus installed in buildings and other structures to minimize their vibrations, which is made up of a mass suspended by one or more damped springs. The oscillation frequency is set close to that of the resonant frequency of the system, minimizing the

amplitude while using significantly less weight compared to the main structure. Igusa and Xu showed that several tuned mass dampers with different frequencies were more efficient in vibration control under wideband excitations than a single tuned mass damper.¹⁰ Mass ratio, frequency ratio, and damping ratio are some of the important factors involved in the design process. The ratio between the frequency of excitation and natural frequency should be greater than the square root of 2.

Methods

Design Principles

The design approach integrated five key principles identified through literature review: (1) Vibration isolation through frequency separation, ensuring the spoon's natural frequencies remain well above the tremor frequency range; (2) Mechanical decoupling between the handle and bowl through a flexible spring element; (3) Optimisation of mass distribution to keep the centre of gravity low; (4) Ergonomics for a comfortable grip; and (5) Bio-mimicry aesthetics for low stigmatisation.

The limitations imposed on the design process were limited to 150 grams weight (130 grams as per literature review is ideal for the elderly population), length less than 250 millimetres, food-grade material, and feasibility of manufacture.

Bio-Inspired Design Elements

Three aspects of nature have been incorporated into this particular spoon design;

Cocoon-Inspired Bowl: The bowl is similar in form to the protective nature of silkworm cocoons. With its deeper curvature, the bowl design has no risk of spillage even under tilted positions. Its multi-layer construction gives it strength, while the inner surface allows effortless

release of food from the spoon. It takes advantage of the protective capabilities of the cocoon that evolved over time to protect contents within its light weight construction.

Bamboo-Inspired Handle: The handle design uses rings that are analogous to nodes on bamboo. In addition to being useful in the way it guides your fingers for gripping the handle for optimal performance, the varied cross-sectional design reflects the stress transfer capabilities of the bamboo. Nodes work like reinforcing regions and give the handle stability, while the hollow portion keeps the weight reduced.

Woodpecker-Inspired Orthogonal Spring: A key feature is the spring that connects the two parts of the spoon. This feature uses a mechanism similar to that used by the woodpecker's hyoid bone. It works as an isolation device to decouple the movement from one part of the spoon to another.

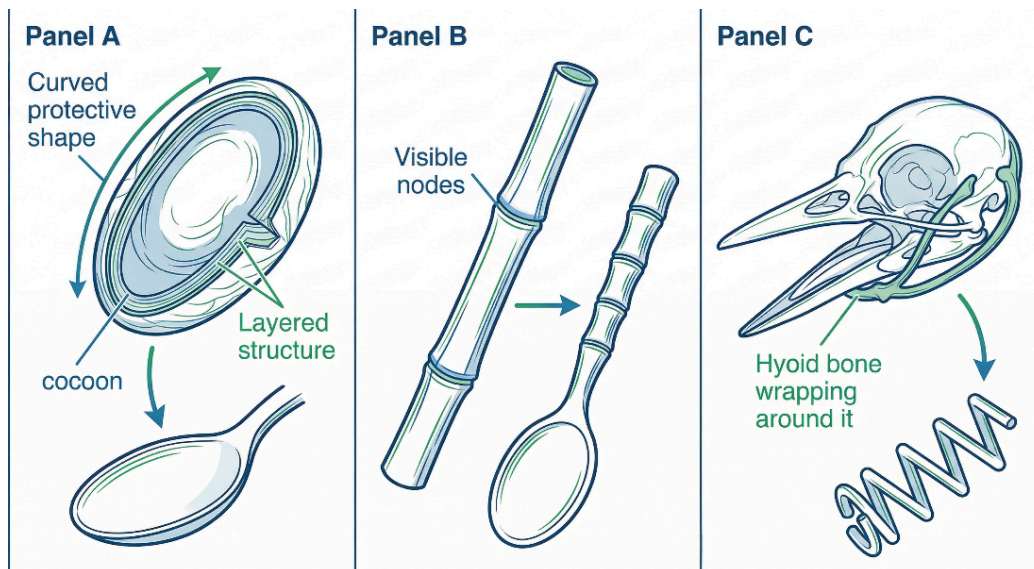


Figure 1. Bio-inspiration sources for the three main design element

CAD Modelling

The spoon was modelled using SolidWorks Student Edition. Overall dimensions are 200 mm, with the bowl having a width of 42 mm, and the handle has a diameter of 19 mm near its grip

end. The orthogonally placed spring has a length of 52 mm between the bowl and handle. The material chosen for construction is AISI 304 stainless steel, which is non-toxic and highly resistant to corrosion, with an elastic modulus of 190 GPa, Poisson's ratio of 0.29, yield strength of 207 MPa, and a density of 8000 kg/m³.

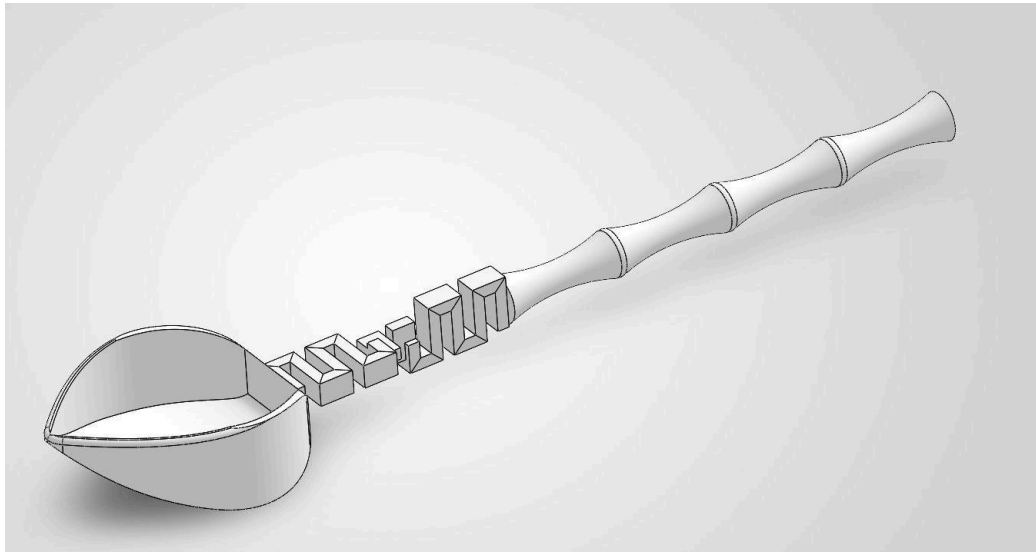


Figure 2. SolidWorks CAD model of the bio-inspired tremor-stabilising spoon.

The mass properties of the final design show that the total mass is 87.89 grams, the volume is 10,985 mm³, and the surface area is 11,570 mm². The center of mass has been calculated to be at $X = -16.48$ mm (closer to the bowl), implying that it is slightly more forward than balanced.

Results and Discussion

Frequency Analysis

The modal analysis method was used to obtain the natural frequency and mode shape of the spoon. The handle was held fixed on one side as the boundary condition, as if the spoon is being held by the user's hand. The natural frequencies obtained for modes 1-5 are shown below: Mode 1 – 58.52 Hz; Mode 2 – 64.16 Hz; Mode 3 – 291.95 Hz; Mode 4 – 497.21 Hz; Mode 5 – 583.97 Hz.

The natural frequency in the first mode, 58.52 Hz, is much higher than the frequency range of 4 to 12 Hz for tremor. This distinction between frequencies is essential in order to achieve vibration isolation. According to the theories dealing with vibrations, when the frequency of the excitation force is significantly lower than the natural frequency of the vibrating object, then the latter behaves as a rigid body. The frequency ratio (tremor frequency / natural frequency), assuming that the frequency of the tremor is maximum at 12 Hz, is 0.2.

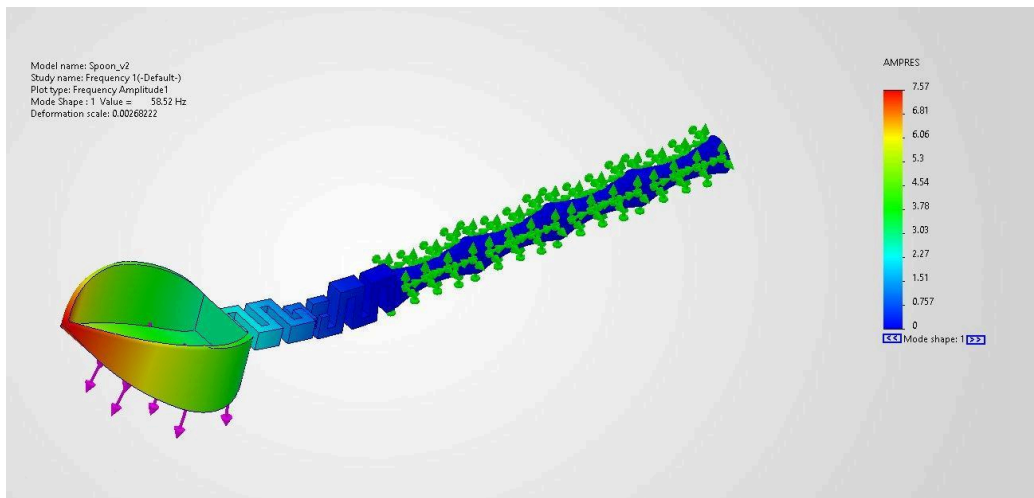


Figure 3. Mode shape 1 at 58.52 Hz showing bending deformation.

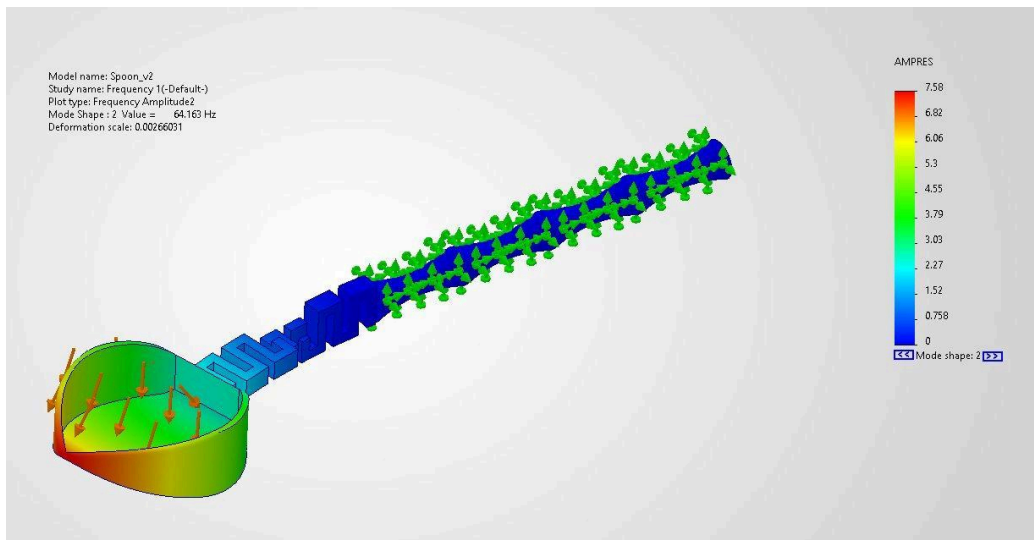


Figure 4. Mode shape 2 at 64.16 Hz showing torsional deformation.

Mode shape analysis indicates that in modes 1 & 2, there is a predominant occurrence of bending and twisting motions in the bowl section in relation to the handle. Orthogonal spring sections display an increased amplitude response, showing their ability to absorb vibration energy. Modes 3, 4, and 5 contain increased deformation of the handle and spring sections.

Static Structural Analysis

Static structure analysis was conducted using the load of 1N at the top surface of the bowl, representing the weight of the food and loads while eating. The other side of the handle was considered a fixed boundary.

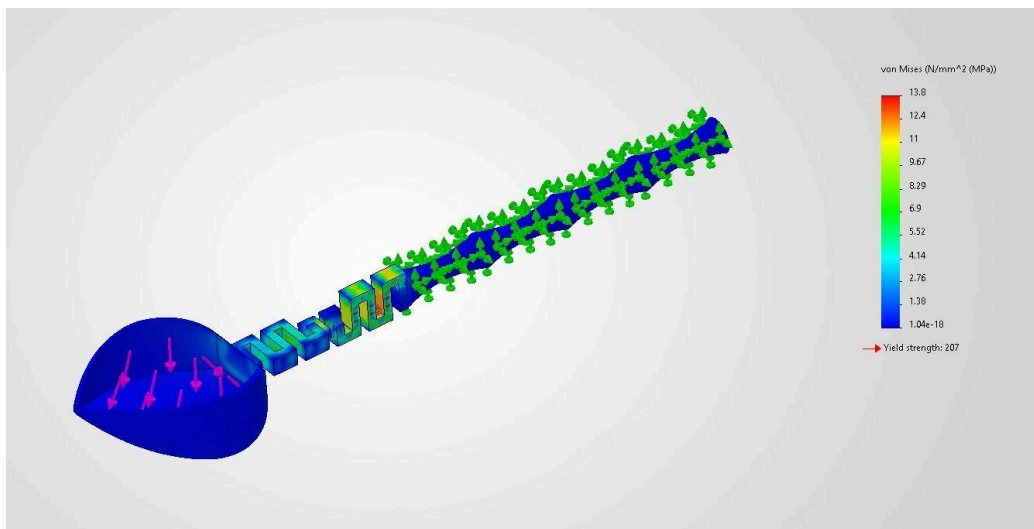


Figure 5. Von Mises stress distribution (maximum 13.8 MPa).

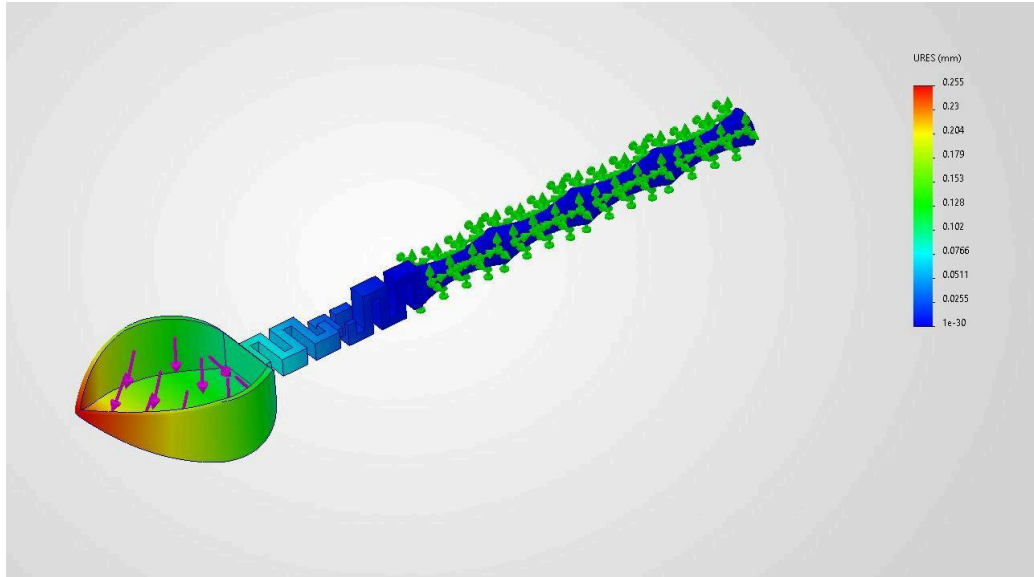


Figure 6. Displacement distribution (maximum 0.255 mm).

The peak von Mises stress of 13.8 MPa is found in the orthogonal spring element, as expected, given that this element experiences the largest strain. This can be compared to the yield strength of AISI 304 stainless steel, which is 207 MPa, resulting in a safety factor of about 15, which represents an extremely conservative design with a large safety margin for increased loading and fatigue factors.

The peak displacement of 0.255 mm at the tip of the bowl is minimal enough for practical purposes. This minimal displacement will ensure that the spoon maintains its structural integrity for its intended purpose of serving food while still offering vibrational damping.

Thermal Analysis

Thermal analysis was carried out to mimic the situation of eating hot food at 125°C, where the temperature of the handle was maintained at room temperature (25°C). Thermal analysis was essential to ensure safety and comfort while consuming hot soups.

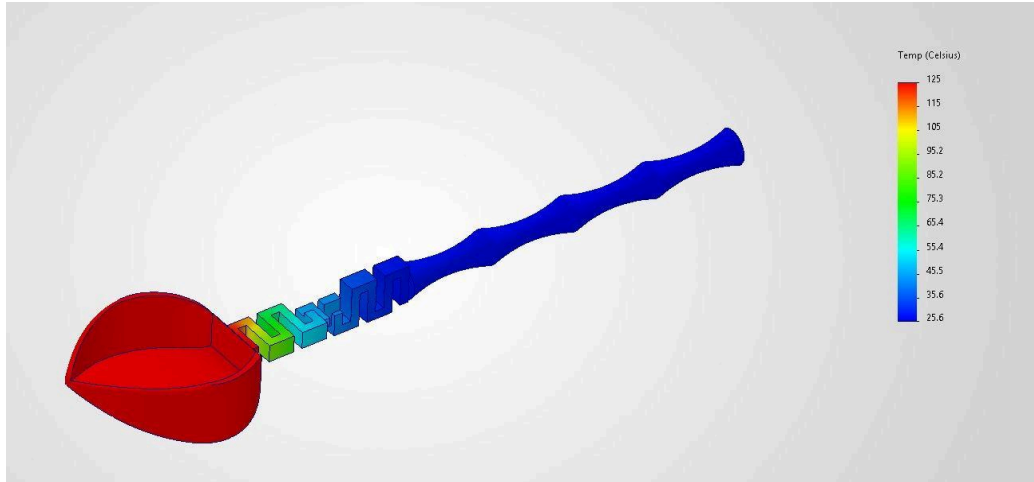


Figure 7. Temperature distribution showing thermal isolation between bowl and handle.

The results reveal that although the temperature inside the bowl is 125°C, it reduces drastically in the orthogonal spring part. The temperature in the handle part stays close to the room temperature, which is around 25 to 35°C. This is because the design of the spring makes the heat transfer route elongated and the cross-section small, which makes the heat transfer difficult. Stainless steel has a lower value of thermal conductivity, which is 16 W/m·K.

Table 1. Summary of simulation results.

S. No.	Parameter	Value	Target/ Limit	Status
1	Total Mass	87.89 g	< 150 g	PASS
2	First Natural Frequency	58.52 Hz	> 12 Hz	PASS
3	Max von Mises Stress	13.8 MPa	< 207 MPa	PASS
4	Safety Factor	15	> 2	PASS
5	Max Displacement	0.255 mm	< 5 mm	PASS

S. No.	Parameter	Value	Target/ Limit	Status
6	Handle Temperature	~300C	< 450C	PASS

Discussion

The results obtained during simulation demonstrate the effectiveness of the bio-inspired approach to the design of an orthodontic device. The key outcome is that the use of an orthogonal spring section ensures proper mechanical decoupling between the handle and the bowl, thus providing effective tremor isolation. As a result of making sure that the first natural frequency (58.52 Hz) is five times greater than the tremor maximum frequency (12 Hz), the danger of the resonance effect that can amplify tremor is eliminated completely. The bowl shape inspired by cocoons ensures spillage prevention due to the curved bottom. The bamboo-inspired handle sections create comfortable grip positions for the user, which will contribute to steadier utensil control. Woodpecker-inspired springs are the main innovation that provides passive vibration isolation. In comparison with the existing solutions, there are several benefits of the current bio-inspired design. For example, unlike weighted utensils, which, according to Meshack and Norman, can cause discomfort due to the fatigue experienced,⁸ the present design offers a passive solution based on the strategic use of flexibility. Also, in contrast to electronic utensils, such as the Liftware Steady system, the proposed design does not require power sources and battery charging. The manufacturing costs calculated using the estimates for metal injection molding and conventional machining are likely to be significantly lower than in electronic utensils, thereby increasing their accessibility for patients in low-resource settings. There are, however, several weaknesses that should be considered. First, at the moment, the paper includes simulation results only, and the physical prototype tests with actual tremor patients are necessary.

Moreover, the design targets only the 4 to 12 Hz tremor frequency range. It means that other frequency ranges will require a different design solution. Furthermore, although the flexure of the spring section allows for more effective vibration decoupling, it might affect the spoon stiffness perceived by the user. Finally, the aesthetic design inspired by nature, while unique, might look strange compared to conventional utensils. The issue is relevant since patients often suffer stigma because of their unusual appearance, as noted by McCoid.¹¹ Future studies should include fabrication of physical prototypes and user testing in order to determine the actual tremor isolation percentage. Experiments involving food items, particularly with the analysis of the difference between wet and dry foods (as observed by Rosa et al.¹²) will provide practical recommendations for patients. Possible material optimization measures involve the search for titanium alloys and silicone coating.

Conclusion

This work shows the possibility of using bio-inspired mechanical solutions in the case of stabilising tremors in eating utensils. Using biological examples such as the silk cocoon, bamboo joint, and woodpecker hyoid bone, a new spoon concept was proposed with a natural vibration damping mechanism provided by an orthogonal spring solution. Simulations have confirmed that the design provides sufficient frequency gap from common tremor frequencies, has sufficient rigidity with a safety factor of 15, acts as insulation for heating hot food up to temperatures of 600 degrees Celsius, and provides an adequate weight of 87.89 grams for elderly patients.

Mechanical design, although less advanced in performance than an electronic alternative, possesses some benefits in terms of costs, durability, and ease of manufacturing. Although experimental studies need to be conducted to test the ability of the device to damp tremors, it is a significant step towards successful implementation.

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